Chapter (25):- Electric Potential

25.1 Electric potential and potential difference:-

* for q_o (test charge) placed in $\vec{E} \Rightarrow \vec{F} = q_o\vec{E}$
  ↳ by some source charge distribution

dw is the work done by this force

$$dw = \vec{F} \cdot d\vec{s} = q_o\vec{E} \cdot d\vec{s}$$

where $d\vec{s}$ is infinitismal displacement.

* the work done by a conservative force = -ve of the change in potential energy (dU)

$$dU = -q_o \vec{E} \cdot d\vec{s}$$

* the change in potential energy for finite displacement of the charge from point A to B

$$\Delta U = -q_o \int_A^B \vec{E} \cdot d\vec{s}$$

, this integral called line or path integral, because $q\vec{E}$ is conservative force, this integral doesint

depend on the path taken from A to B.

Electric potential (V):- the potential energy Per unit charge.

$$V = \frac{U}{q_o}$$ ⇒ Electric potential

Electric potential difference (ΔV)

$$\Delta V = V_B - V_A$$

$$\therefore \Delta V = \frac{\Delta U}{q_o} = -\int_A^B \vec{E} \cdot d\vec{s}$$ ⇒ potential difference between two points

Electric potential equal zero in infinity

$$V_\infty = 0$$

if an external agent moves a test charge from A to B without changing K.E for the test charge ⇒ the agent perform a work $W = \Delta U = q \Delta V$

$\Delta V \equiv [\text{volt}]$

1 volt = 1 J/C

Volt :- 1 J work must be done to move 1 C charge through potential difference = 1 volt

$E \equiv [N/C] \equiv [V/m]$

We can introduce $\vec{E}$ as a measure of the rate of change with position.

Potential

a unit of energy used in atomic and nuclear physics is electron volt (e.v).

electron volt :- is the energy gain or lose when a charge of magnitude e^- is moved through potential difference (1 volt).

$\boxed{1 \text{ e.v}} = 1.6 \times 10^{-19} \text{ C} \times 1 \text{ volt}$

$= 1.6 \times 10^{-19} \text{ C.volt} = \boxed{1.6 \times 10^{-19} \text{ J}}$

Example:- Calculate the work needed to move a +ve charge from A to B, where

$V_A = 40$ volt , $V_B = 60$ volt , $q = 4\mu C$

$$\Delta V = V_B - V_A = \frac{W}{q}$$

$$W = q\,(V_B - V_A)$$

$$= 4 \times 10^{-6}\,(60 - 40)$$

$$= 80 \times 10^{-6}\ J$$

$$= \frac{80 \times 10^{-6}}{1.6 \times 10^{-19}} = 5 \times 10^{14}\ e.V$$

25.2 Potential difference in a uniform electric field:-

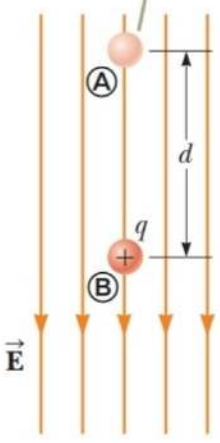
$$\Delta V = V_B - V_A = -\int_A^B \vec{E} \cdot d\vec{s}$$

$$= -\int_A^B E \cos(0)\, ds$$

$$= -\int_A^B E\, ds = -E \int_A^B ds$$

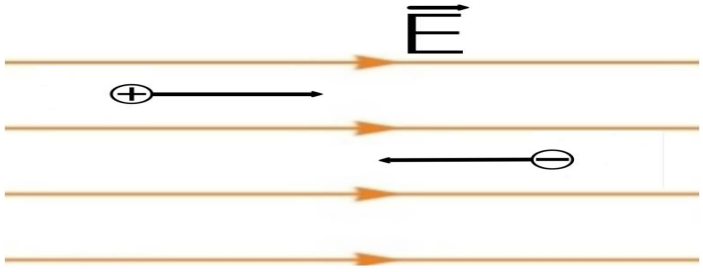
$$\boxed{\Delta V = -E d}$$

When a positive charge moves from point Ⓐ to point Ⓑ, the electric potential energy of the charge–field system decreases.



-ve Potential difference $\Rightarrow V_A > V_B$

Electric field lines always point in the direction of decreasing electric potential.



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$$\Delta U = q\Delta V = -qEd$$

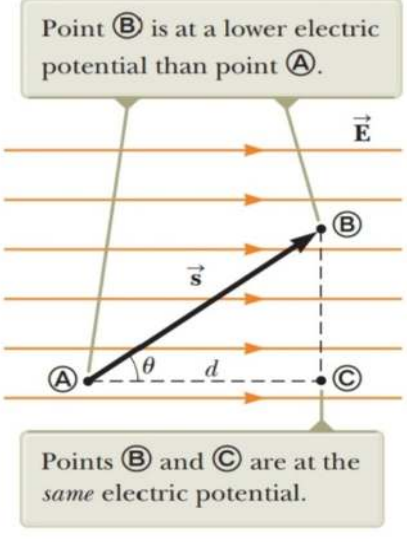
* for +ve charge ΔU decreases
* for -ve charge ΔU increases.

in general a charged particle moves between A and B in uniform $\vec{E}$ such that $\vec{s}$ is not parallel to the field lines

$$\Delta V = -\int_A^B \vec{E}\cdot d\vec{s}$$

$$= -\vec{E}\cdot\vec{s}$$

$$\Delta U = q_0\Delta V = -q_0\vec{E}\cdot\vec{s}$$

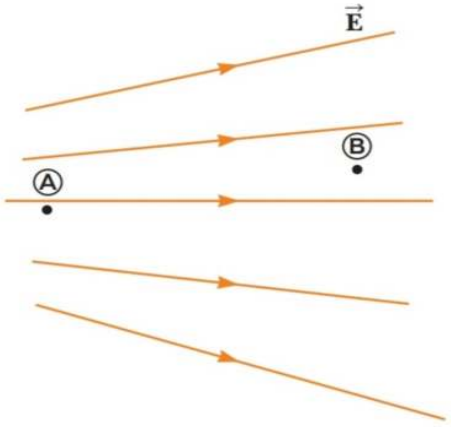


* All points in a plane ⊥ to $\vec{E}$ are at the same electric potential (equipotential surfaces).

Quick Quiz 25.1 In Figure 25.1, two points A and B are located within a region in which there is an electric field. (i) How would you describe the potential difference $\Delta V = V_B - V_A$? (a) It is positive. (b) It is negative. (c) It is zero. (ii) A negative charge is placed at A and then moved to B. How would you describe the change in potential energy of the charge–field system for this process? Choose from the same possibilities.

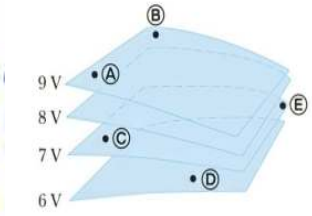
i-(b) it's negative.

ii-(a) it's positive.



$$\Delta U = -q\vec{E}\cdot\vec{s}$$
$$= -(-q)\vec{E}\cdot\vec{s}$$
$$= q\vec{E}\cdot\vec{s} \quad +ve.$$

Quick Quiz 25.2 The labeled points in Figure 25.4 are on a series of equipotential surfaces associated with an electric field. Rank (from greatest to least) the work done by the electric field on a positively charged particle that moves from A to B, from B to C, from C to D, and from D to E.



from A to B

$\Delta U_{AB} = q\Delta V = q(9-9) =$ zero

$\Delta U_{BC} = q(7-9) = -2q$

$\Delta U_{CD} = q(6-7) = -q$

$\Delta U_{DE} = q(7-6) = q$

$W_{BC} > W_{CD} > W_{AB} > W_{DE}$ ⇐ Because $\boxed{W = -\Delta U}$

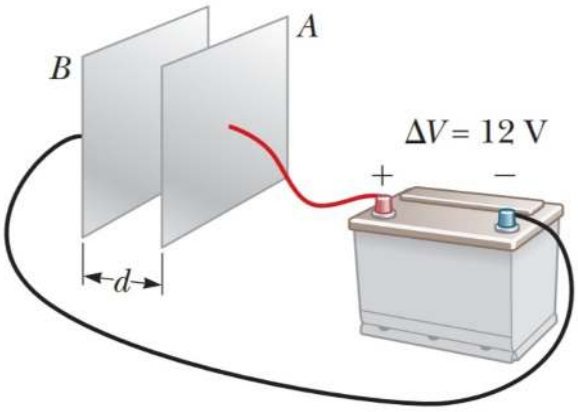
$2q > q >$ zero $> -q$

-6-

Ex(25.1):- $\vec{E}$ between two parallel plate:-

$\Delta V = 12$ volt

$d = 0.3$ cm



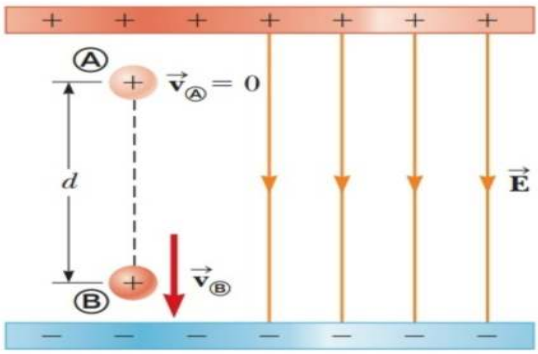
$$E = \frac{\Delta V}{d} = \frac{12}{0.3 \times 10^{-2}}$$

$$= \boxed{4 \times 10^{3} \text{ V/m}}$$

Ex(25.2):- motion of a proton in a uniform $\vec{E}$:-

$E = 8 \times 10^{4}$ V/m

$d = 0.5$ m



$$\Delta V = -Ed$$

$$= \boxed{-8 \times 10^{4} \times 0.5 \text{ V}}$$

$$\Delta V = -4 \times 10^{4} \text{ volt} \Rightarrow V_B < V_A$$

$$\Delta U = q_o \Delta V$$

$$= 1.6 \times 10^{-19} \times -4 \times 10^{4}$$

$$\boxed{\Delta U = -6.4 \times 10^{-15} \text{ J}}$$

decreasing of potential energy corresponding to increasing of Kinetic energy of the charge.

For isolated system

$\Delta K + \Delta U = 0$ (conservation of energy).

$\left(\frac{1}{2}mv^2 - 0\right) + e\Delta V = 0$

$\frac{1}{2}mv^2 = -e\Delta V$

$v^2 = \frac{-2e\Delta V}{m} \Rightarrow v = \sqrt{\frac{-2e\Delta V}{m}}$

$v = \sqrt{\frac{-2e(-Ed)}{m}} = \sqrt{\frac{2eEd}{m}}$

$v = \sqrt{\frac{2 \times 1.6 \times 10^{-19} \times 8 \times 10^{4} \times 0.5}{1.67 \times 10^{-27}}}$

$\boxed{v = 2.8 \times 10^{6}\ m/s}$

25.3 Electric Potential and potential energy due to Point charges:-

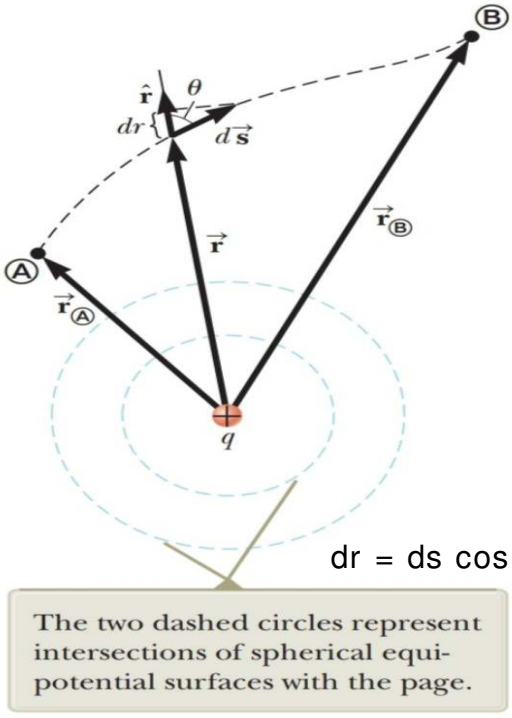
$$V_B - V_A = -\int_A^B E \cdot ds$$

$$\vec{E}\cdot d\vec{s} = \frac{Kq}{r^2}\,dr \quad *$$

$$V_B - V_A = -Kq \int_{r_A}^{r_B} \frac{dr}{r^2}$$

$$= -Kq\left(-\frac{1}{r}\right)_{r_A}^{r_B}$$

$$\Delta V = Kq\left[\frac{1}{r_B} - \frac{1}{r_A}\right]$$



$dr = ds\cos\theta$

The two dashed circles represent intersections of spherical equi-potential surfaces with the page.

if Ⓐ at infinty $r_A = \infty$

$\therefore V_A = 0$

$$V = \frac{Kq}{r}$$ Potential due to point charge.

$$* \vec{E}\cdot d\vec{s} = K_e \frac{q}{r^2}\hat{r}\cdot d\vec{s}$$

$\hat{r}\cdot d\vec{s} = |\hat{r}|\,ds\cos\theta$

$= (1)\,ds\cos\theta$

$dr = ds\cos\theta$ (i.e.)

$\therefore \hat{r}\cdot d\vec{s} = dr$

$\therefore \vec{E}\cdot d\vec{s} = K_e \frac{q}{r^2}\,dr$

* for a group of point charges :-

$$V = K\sum_i \frac{q_i}{r_i}$$

the work needed to bring q_2 from ∞ to P

$$W = q_2 V_P$$

$$= q_2\left(\frac{Kq_1}{r}\right)$$

$$W = \frac{Kq_1q_2}{r} = U$$

V_P

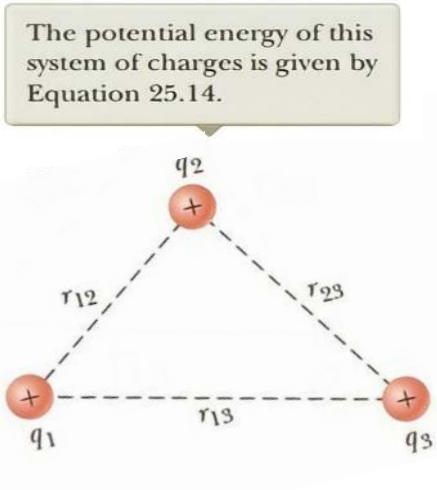
P $\quad \vec{r} \quad$ q_1

for charges of same sign, U is +ve

for charges of opposite sign, U is -ve.

if the system consist of more than 2 charges

$$U = K\left[\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}}\right]$$



in general

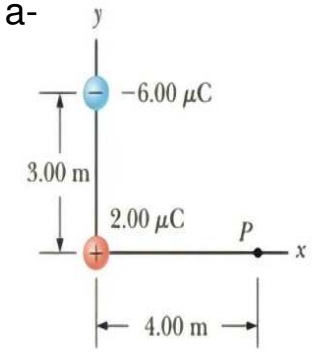
$$U = K \sum_i \sum_j \frac{q_i q_j}{r_{ij}}$$

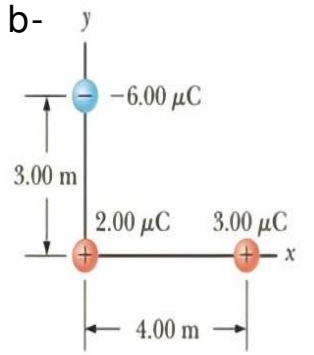
Ex(25.3):- electric potential due to 2-point charges:-

a- $V_p = K\left(\frac{q_1}{r_1} + \frac{q_2}{r_2}\right)$

$$V_p = 9\times10^9\left(\frac{2\times10^{-6}}{4} - \frac{6\times10^{-6}}{5}\right)$$

$$= \boxed{-6.3\times10^3 \text{ volt}}$$





b- $\Delta U = U_f - U_i = U_f - \cancel{U_\infty}^{0} = U_f$

$$U_f = q_3 V_p = 3\times10^{-6} \times -6.3\times10^3$$

$$= \boxed{-1.9\times10^{-2} \text{ J}}$$

$W = -U = +1.9\times10^{-2}$ J to take q_3 back from p to ∞.

25.4 Obtaining the value of $\vec{E}$ from the value of V

$$\Delta V = -\int \vec{E} \cdot d\vec{s}$$

$$dV = -\vec{E} \cdot d\vec{s}$$

$$d\vec{s} = d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$$

$$\vec{E} = -\frac{dV}{d\vec{s}}$$

$$E_x = -\frac{dV}{dx} \quad , \quad E_y = -\frac{dV}{dy} \quad , \quad E_z = -\frac{dV}{dz}$$

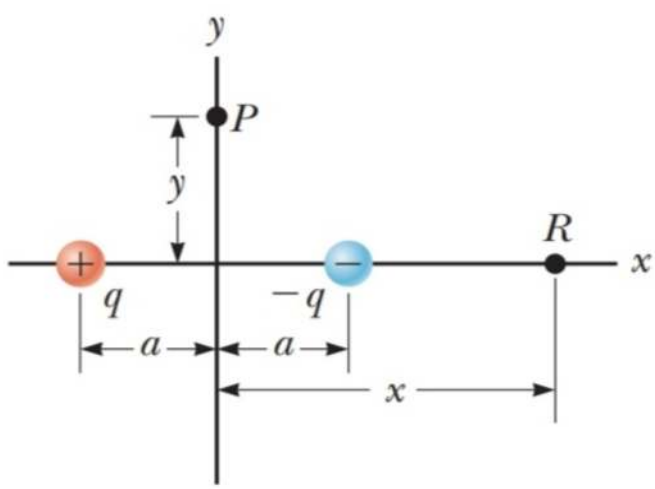
$$\vec{E} = E_x\hat{i} + E_y\hat{j} + E_z\hat{k}$$

$$= -\left[\frac{dV}{dx}\hat{i} + \frac{dV}{dy}\hat{j} + \frac{dV}{dz}\hat{k}\right]$$

EX. (25.4) Electric potential due to a dipole :-

a- $$V_P = K\sum_i \frac{q_i}{r_i}$$

$$V_P = \left[\frac{Kq}{\sqrt{a^2+y^2}} - \frac{Kq}{\sqrt{a^2+y^2}}\right] = 0$$



b- $$V_R = K\left[\frac{-q}{x-a} + \frac{q}{x+a}\right]$$

$$= Kq\left[\frac{1}{x+a} - \frac{1}{x-a}\right]$$

$$V_R = Kq\left[\frac{x-a-x-a}{x^2-xa+xa-a^2}\right]$$

$$\boxed{V_R = \frac{-2Kqa}{x^2-a^2}}$$

c- for $x \gg a$

$$V = \lim_{x \gg a}\left(\frac{-2Kqa}{x^2-a^2}\right) \approx \boxed{\frac{-2Kqa}{x^2}}$$

$$E_x = -\frac{dV}{dx} = -\frac{d}{dx}\left(\frac{-2Kqa}{x^2}\right)$$

$$= 2Kqa\,\frac{d}{dx}\left(\frac{1}{x^2}\right)$$

$$\boxed{E_x = \frac{-4Kqa}{x^3}}$$

Example:- let $V = a_o(5x^2y + 3xyz - 2yz^2)$

a_o is constant $= 4\times10^{-2}$, find E at (2, -1, 1)

$$E_x = -\frac{dV}{dx} = -a_o(10xy + 3yz) = \boxed{+0.92\ V/m}$$

$$E_y = -\frac{dV}{dy} = -a_o(5x^2 + 3xz - 2z^2) = \boxed{-0.96\ V/m}$$

$$E_z = -\frac{\partial V}{\partial z} = -a_0(3xy - 4yz) = 0.08 \text{ V/m}$$

$$\vec{E} = E_x\hat{i} + E_y\hat{j} + E_z\hat{k}$$

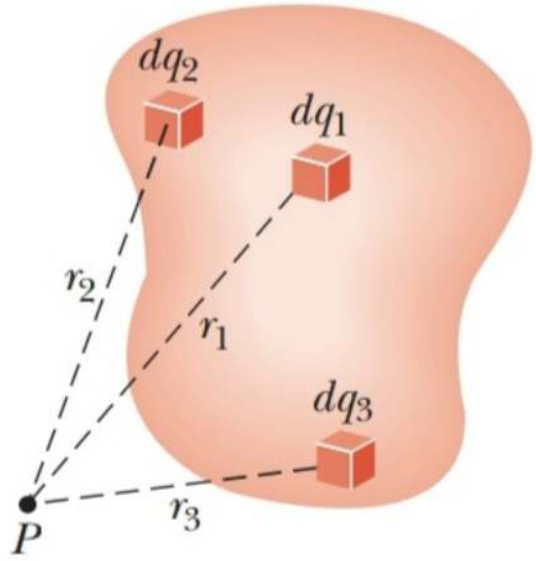
$$= +0.92\,\hat{i} - 0.96\,\hat{j} + 0.08\,\hat{k} \quad (\text{V/m})$$

25.5 Electric potential due to continous charge distribution:-

Electric potential due to charge element dq is

$$dv = \frac{K\,dq}{r} \Rightarrow$$

$$\boxed{V_p = \int \frac{K\,dq}{r}}$$

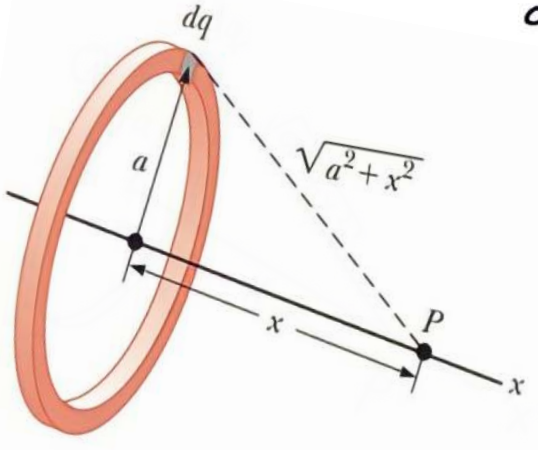


remember that $\Rightarrow \lambda = \frac{Q}{L} \equiv C/m \Rightarrow dq = \lambda dL$

$$\sigma = \frac{Q}{A} \equiv C/m^2 \Rightarrow dq = \sigma dA$$

$$\rho = \frac{Q}{V} \equiv C/m^3 \Rightarrow dq = \rho dv$$

Ex(25.5):- Electric potential due to a uniformly charged ring:-



a- $V_p = K\int \frac{dq}{r} = K\int_0^Q \frac{dq}{\sqrt{a^2+x^2}}$

$$= \frac{Kq}{\sqrt{a^2+x^2}}\Big]_0^Q$$

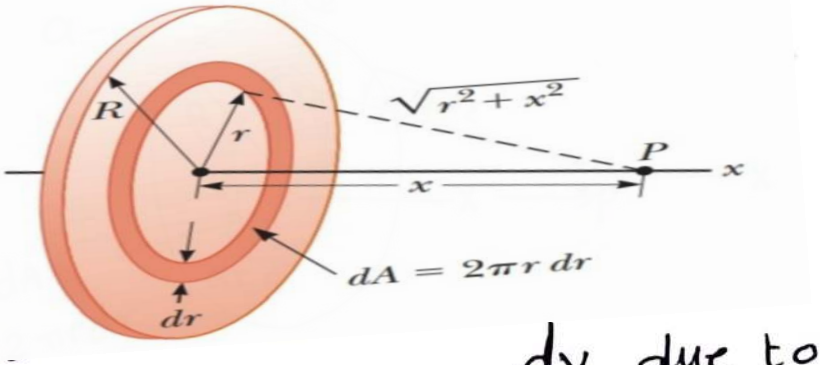
$$\boxed{V_p = \frac{KQ}{\sqrt{a^2+x^2}}}$$

b- $E_x = -\frac{dV}{dx} = -KQ\frac{d}{dx}(a^2+x^2)^{-1/2}$

$= -KQ\left(-\frac{1}{2}\right)(a^2+x^2)^{-3/2}(2x)$

$$E_x = \frac{K \times Q}{(a^2+x^2)^{3/2}}$$

Ex(25.6):- Electric Potential due to a uniformly charged disk.



$dq = \sigma\, dA$

$= \sigma(2\pi r\, dr)$

$= 2\pi\sigma r\, dr$

dv due to ring from previous example

$$dv = \frac{K\,dq}{\sqrt{r^2+x^2}} = \frac{K\,2\pi\sigma r\,dr}{\sqrt{r^2+x^2}}$$

$$V_p = K\pi\sigma \int_0^R \frac{2r\,dr}{\sqrt{r^2+x^2}}$$

let $U = r^2 + x^2 \Rightarrow dU = 2r\,dr$

$$\int \frac{2r\,dr}{\sqrt{r^2+x^2}} = \int \frac{dU}{\sqrt{U}} = 2\sqrt{U} = 2\sqrt{r^2+x^2}\Big]_0^R$$

$$= 2\left[\sqrt{R^2+x^2} - x\right]$$

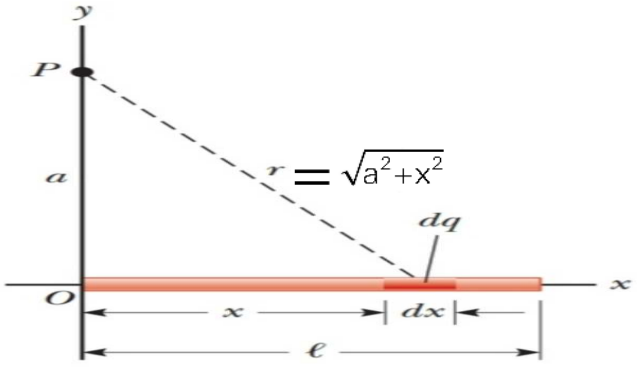
$$V_p = 2\pi\sigma K\left[\sqrt{R^2+x^2} - x\right]$$

b- $E_x = -\frac{dV}{dx} = -2\pi K\sigma \frac{d}{dx}\left[\sqrt{R^2+x^2} - x\right]$ ($\uparrow (R^2+x^2)^{\frac{1}{2}}$)

$$= -2\pi K\sigma \left[\frac{1}{2} \cdot \frac{2x}{\sqrt{R^2+x^2}} - 1\right]$$

$$\boxed{E_x = 2\pi K\sigma \left[1 - \frac{x}{\sqrt{R^2+x^2}}\right]}$$

Ex (25.7):- Electric Potential due to a finite Line of charge:-



$$dV = K\frac{dq}{r} = \frac{K\lambda dx}{\sqrt{a^2+x^2}}$$

$$V_p = K\lambda \int_0^L \frac{dx}{\sqrt{a^2+x^2}}$$

$$\int \frac{dx}{\sqrt{a^2 \pm x^2}} = \ln\left|x \pm \sqrt{a^2 \pm x^2}\right|$$

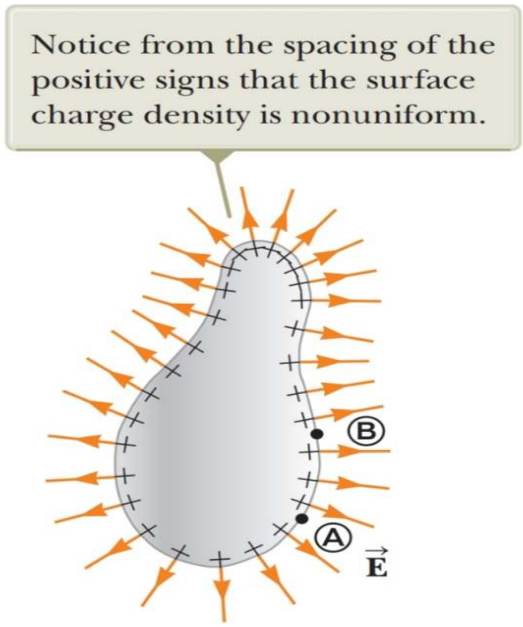
$$V_p = K\lambda \ln\left|x + \sqrt{a^2+x^2}\right|_0^L$$

$$= K\lambda\left[\ln\left(L+\sqrt{a^2+L^2}\right) - \ln a\right]$$

$$\boxed{V_p = K\lambda\left[\ln\left(\frac{L+\sqrt{a^2+L^2}}{a}\right)\right]}$$

25.6 Electric Potential due to a charged conductor:-

We Know that for solid charged conductor in equalibrium charge resides on the outer surface of conductor, so Electric field inside it = zero and $\vec{E}$ just outside the conductor is ⊥ to the surface.



$\vec{E}$ always ⊥ $d\vec{s}$

$\therefore \boxed{\vec{E} \cdot d\vec{s} = 0}$

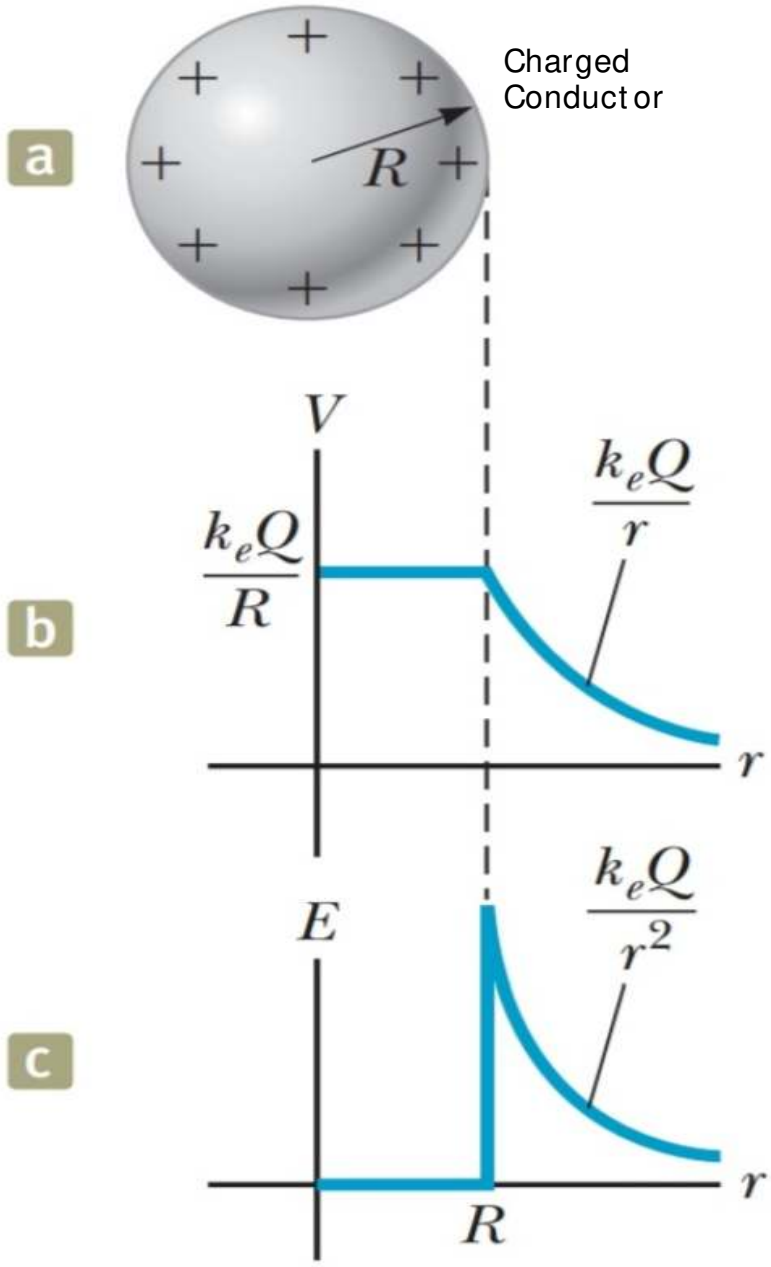
$* \Delta V = V_B - V_A = -\int_A^B \vec{E} \cdot d\vec{s}$

$= 0$

$\therefore V_B = V_A \Rightarrow$ ① V is constant every where on the surface of the charged conductor.

② the surface of any charged conductor in electro-static equalibrium is an equipotential surface.

③ because $\vec{E}$ = zero inside conductor; the electric Potential is constant inside it and equal it's value on the surface.


a
Charged
Conductor
R
b
V
$\frac{k_e Q}{R}$
$\frac{k_e Q}{r}$
r
c
E
$\frac{k_e Q}{r^2}$
r
R

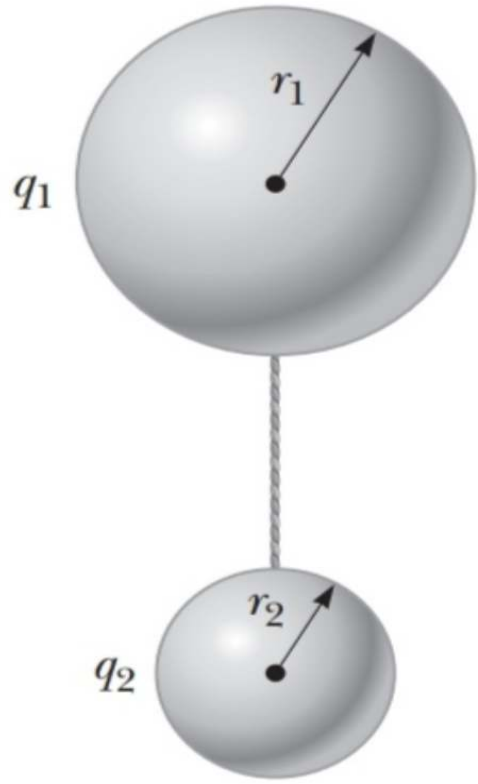
EX(25.8):- Two connected charged spheres:-

for connected spheres

$$V_1 = V_2$$

$$\frac{K q_1}{r_1} = \frac{K q_2}{r_2}$$

$$\therefore \boxed{\frac{q_1}{q_2} = \frac{r_1}{r_2}}$$



$$E_1 = \frac{K q_1}{r_1^2} \quad , E_2 = \frac{K q_2}{r_2^2}$$

$$\frac{E_1}{E_2} = \frac{\frac{K q_1}{r_1^2}}{\frac{K q_2}{r_2^2}}$$

$$= \frac{\frac{q_1}{r_1^2}}{\frac{q_2}{r_2^2}} = \frac{q_1}{r_1^2} \cdot \frac{r_2^2}{q_2}$$

$$= \frac{q_1}{q_2} \cdot \frac{r_2^2}{r_1^2} = \frac{r_1}{r_2} \cdot \frac{r_2^2}{r_1^2}$$

$$\boxed{\frac{E_1}{E_2} = \frac{r_2}{r_1}} \Rightarrow \boxed{E_1 = E_2 \frac{r_2}{r_1}}$$

$\therefore V_1 = V_2$

$E_2 > E_1$